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J.M. Romero-García, C. Martínez-Patiño, E. Ruiz, I. Romero, E. Castro*

Ethanol production from olive stone hydrolysates by xylose fermenting microorganisms

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Abstract: Olive stones are the main solid byproducts obtained from olive oil production and from table olives production. As a lignocellulosic material, the use of olive stones for ethanol and other chemicals production has been proposed, particularly under the biorefinery concept. As part of such a process, this work deals with the fractionation of the lignocellulosic material by dilute acid autoclave pretreatment at 2% sulfuric acid, 130°C, 60 min and 1:1 liquid to solid ratio. Moreover, the work addresses the fermentation of the liquors obtained after pretreatment. The released sugars are composed mainly by xylose and other hemicellulosic sugars. The fermentation performance of three xylose-fermenting microorganisms, e.g. two *Escherichia coli* species and *Scheffersomyces stipitis*, are compared. The study analyzes in a first step the microorganism behavior on synthetic liquors, with a similar composition to that of the real liquors. Finally, and taken into account the results from the previous steps, the real liquor obtained from olive stones pretreatment is fermented. Results show that *E. coli* MM160 is the best ethanol producer out of the three microorganisms studied. Globally, the pretreatment produced a liquor containing 140 g hemicellulosic sugars/l and requiring firstly dilution by 50% and a detoxification step by overliming. The fermentation of this liquor by *E. coli* MM160 results in a 25 g ethanol/l solution equivalent to 50 g ethanol/kg olive stone, in spite of 20 g acetic acid/l also present. These results confirm both olive stones and *E. coli* MM160 as promising feedstock and microorganism for ethanol production.

Keywords: Olive stones, bioethanol, biorefinery, xylose fermenting microorganisms, ethanologenic *Escherichia coli, Scheffersomyces stipitis*

1 Introduction

Olive oil, traditionally produced in the Mediterranean countries, is gaining interest worldwide due to its consideration as a healthy fat. Thus, olive tree cultivation has been spread and nowadays more than nine million hectares of olive trees are cultivated all over the world. According to FAOSTAT, the average annual production of olives in the period 2006-2012 was over 18 million tonnes [1]. Approximately 10% by weight of the olive fruit is the olive stone, which is crushed along with the rest of the fruit during the extraction process and is recovered after oil separation. For a detailed process diagram, see [2].

Although different applications like adsorbents, source of polyols or raw material for activated carbon production have been proposed [3], the main use for olive stones at the present time is heat production, due to a relatively high heating value of 20 MJ/kg [4].

Olive stones are constituted mainly by cellulose (24.1-40.4%), hemicelluloses (18.5-34.4%) and lignin (25.3-32.5%) [2]. Considering the sugar content of olive stone, its use as raw material for ethanol production has been proposed. The conversion process includes a pretreatment, enzymatic hydrolysis and fermentation as the main steps. Depending on the pretreatment conditions, the hemicellulose fraction may be partially dissolved, as well as other lignin-derived compounds, rendering the solid lignocellulosic matrix more accessible to the enzymatic hydrolysis which is the next step of the process. Finally, the conversion of the released glucose is accomplished by yeast fermentation. However, for taking full advantage of the hemicellulosic sugars solubilized during the pretreatment (mainly xylose), which can account for up to 50% of total available sugars, the use of microorganisms with the ability of ferment also pentoses is required.

***Corresponding author E. Castro,** Dept. Chemical, Environmental and Materials Engineering, Agrifood Campus of International Excellence, ceiA3, Universidad de Jaén, 23071 Jaén, Spain, E-mail: ecastro@ujaen.es
J.M. Romero-García, C. Martínez-Patiño, E. Ruiz, I. Romero, Dept. Chemical, Environmental and Materials Engineering, Agrifood Campus of International Excellence, ceiA3, Universidad de Jaén, 23071 Jaén, Spain

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Among the natural occurring pentose-fermenting yeasts, *Scheffersomyces (Pichia) stipitis* has been described to provide high ethanol yields from C5 sugars. In addition, its low nutrient requirements along with a high resistance to contaminants with a potential cleanup of some toxins have been reported to claim this yeast as the most promising one for pentose fermentation [5]. Using olive-derived hydrolysates, *S. stipitis* has been used as fermenting microorganism for olive tree pruning biomass conversion [6,7].

The low performance of yeasts when fermenting pentoses compared to results from glucose [5] and the effect on inhibitory compounds which make necessary a detoxification procedure [8] are usually mentioned as the main drawbacks of using natural-occurring yeasts for ethanol production from pentose hydrolysates.

The most commonly studied inhibitors found in lignocellulosic hydrolysates are weak acids (acetic acid, formic acid and levulinic acid), furan derivatives (furfural and hydroxymethylfurfural) and phenolic compounds (solubles from the partial degradation of lignin and others). Other factors may also cause inhibition, e.g., the presence of inorganic salts, high sugar concentrations and the ethanol produced in the fermentation [8,9]. The so-called overliming or treatment with calcium hydroxide is one of the preferred methods for efficiently removing or reducing the concentration of inhibitory compounds. When applied on sulphuric acid derived hydrolysates, overliming results in the precipitation of gypsum, calcium sulfate, contributing to maintain a low concentration of soluble salts, which is favourable for the fermentation process [8].

In an attempt to overcome these issues, a number of studies have focused on the modification of other microorganisms, by different techniques including direct evolution, genetic modification, gene encoding and so on. *Escherichia coli* is one of the most widely studied microorganisms in such a way that a wide range of products are obtained at industrial scale [9]. *E. coli* KO11 was first reported in 1991 as an ethanologenic strain from pentoses. Since then, improving results have been reported, based on different substrates, and exhibiting a remarkable stability with steady ethanol yield for more than 26 days of cultivation at dilution rates values of 0.045 and 0.075 h^{-1} [9].

Using lignocellulosic materials as sugar source, ethanologenic *E. coli* has also been applied to pentose fermentation on sugarcane bagasse [10], Eucalyptus [11], Pinus sp. [9] or corn stover [12], among others. Related to olive-derived biomass, little information is available on the use of ethanologenic *E. coli*, and only a study on olive pomace hydrolysates is available [13].

The aim of this work was to compare the performance of several xylose fermenting microorganisms on hydrolysates (liquid fractions) obtained from pretreatment of olive stones. The comparison of the microorganism behavior is established firstly on synthetic liquors, e.g. solutions with similar composition to that of the pretreatment, and then the resulting findings are applied to the fermentation of the real olive stone hydrolysate. The mass balance of the process will also be calculated and discussed.

2 Materials and Methods

2.1 Raw material

Olive stones were kindly supplied by a local olive oil factory. The stones were separately removed from the olive pomace with an industrial pitting machine, with a 6 mm sieve separator, which is the standard size in this industrial process, soaked in water, washed to free them from any adhering flesh, air-dried and then dried for 24 h at 50°C (moisture next to 6% w). The composition (% dry matter ± standard deviation) includes cellulose (24.1±1.3), hemicelluloses (34.4±1.1, in which xylose 28.8±0.9), lignin (32.5±0.3), extractives (8.9±0.1) and ash (0.5±0.01).

2.2 Pretreatment

As the main interest of the present work was to establish the performance of several xylose fermenting microorganisms, the pretreatment conditions were selected in such a way that the hemicellulosic sugars were the target of the process. Based on our previous experience, a pretreatment in autoclave by dilute sulfuric acid (2% w/v) at 130°C and a solid-liquid ratio 1:1 (w/w, 150 g of olive stone) was performed. The pretreatment procedure consisted of two steps of 1 h, in which two lots of fresh solvent was used on the same solid. Then the resulting liquors or hydrolysates were mixed together. This process resulted in a highly concentrated hemicellulosic sugar solution, while the solid is concentrated in lignocellulose matrix due to hemicellulosic solubilisation. Both fractions were separated by filtration, and the sugar solution was further used as fermentation broth.

The pretreated solid, in which the hemicellulosic fraction has been substantially reduced, may be used for glucose production by a further, more severe pretreatment step followed by enzymatic hydrolysis. This will be the focus of future works.

2.3 Detoxification

An overliming procedure based on the reports by Martínez et al. [14] was applied to the liquors. It consisted in adding $Ca(OH)_2$ until pH 10. Then the liquor was agitated in an orbital shaker for 30 min at 50°C and 150 rpm and finally it was centrifuged at 3,500 rpm for 10 min for solid separation (Rotina 420, Hettich Zentrifugen, Germany).

2.4 Microorganisms

Three microorganisms with contrasted ability to ferment pentose solutions were selected to compare performance on olive stone hydrolysates. These were *Scheffersomyces stipitis* CBS 6054, *Escherichia coli* strain MS04 and *Escherichia coli* MM160.The ethanologenic *E. coli* strain MS04 was kindly donated by Dr. Martínez from Institute of Biotechnology (UNAM, Mexico) while *E. coli* MM160 was kindly provided by Dr. Ingram from University of Florida, USA.

S. stipitis was maintained in agar slants and transferred prior to the fermentation into 75 mL culture medium composed of 5 g/l yeast extract, 1 g/l NH_4Cl, 1 g/l KH_2PO_4, 0,5 g/l $MgSO_4 \cdot 7H_2O$ and 40 g/l xylose. The culture medium was sterilized in autoclave at 121°C for 20 min and then inoculated and grown at 30°C in an orbital incubator at 180 rpm for 24 h.

E. coli MS04 was maintained in glycerol tubes at -80°C and transferred before inoculation to modified AM1 culture medium [10,15] which was previously sterilized by filtration through 0.2 µm membranes. Finally the inoculum was grown at 37°C in an orbital incubator at 180 rpm for 24 h. A similar procedure was followed for *E. coli* MM160.

2.5 Fermentation experiments

All fermentation experiments were performed with 0.5 g/l initial concentration of microorganism (dry matter). Two types of fermentations were assayed, depending on the pH control of the fermentation broths. For uncontrolled cultures, fermentations were performed in 250 ml cotton-stopped Erlenmeyer flasks using 75 ml inoculated liquor. Temperature control and agitation were provided by using an orbital incubator. Samples were withdrawn at different times in a laminar flow cabin (TELSTAR mod. AH-100, Spain).

pH-controlled cultures were studied in 300 ml glass flasks, with 150 ml fermentation liquor provided with pH probe and agitation by magnetic spin. The temperature control is assured by using a water bath, and pH is monitored and automatically corrected by addition of 2M KOH. Each flask was stopped by a robber cap, where three holes allowed for sampling, pH monitoring and venting.

2.6 Analytical methods

The composition of raw material was determined according to NREL (National Renewable Energy Laboratory, [16]) analytical methods for biomass. Liquors obtained from autoclave were centrifuged and filtered through 0.45 µm membranes (Gelman Sciences Inc., Michigan, USA) and analyzed by HPLC for quantitative carbohydrate analysis. The HPLC system (Waters, Milford, USA) was equipped with a refractive index detector (model 2414). A CARBOSep CHO-782 Pb (Transgenomic, Inc., Omaha, USA) carbohydrate analysis column operating at 70°C with ultrapure water as a mobile-phase (0.6 mL/min) was used for the monomeric sugars (glucose, xylose, galactose, arabinose and mannose) determinations. Furfural, hydroxymethylfurfural (HMF), acetic acid, formic acid and ethanol content were analyzed by HPLC in a Hewlett-Packard 1100 system (Palo Alto, CA, USA) equipped with a refractive index detector. The separation was performed with a Aminex HPX-87H column (Bio-Rad, Hercules, CA, USA) operating at 65°C with 5 mM sulfuric acid as an eluent at a flow rate of 0.6 mL/min.

3 Results and Discussion

3.1 Pretreatment

The results from the pretreatment with dilute sulfuric acid in the autoclave are shown in Table 1. As can be seen, the liquors obtained in the two steps contained sugar concentrations (glucose, xylose, galactose and arabinose) quite above 100 g/l, which could result in high ethanol concentration following fermentation with appropriate microorganisms. Hemicellulosic sugars (xylose, galactose and arabinose) represent over 90% of the sugars present in the liquor and within these xylose represents over 80%. These pretreatment conditions gave also low degradation levels of formic acid, furfural and HMF, and high levels of acetic acid. Nevertheless, the concentrations of both furfural and HMF are higher in the liquors obtained from the second stage of the pretreatment, e.g., liquid phases from the second sulfuric acid solution. This can be attributed to degradation of the sugars remaining in the solid issued from the first acid contact.

Table 1. Results of pretreatment with dilute acid (in triplicate, error <5%).

Liquor	1st Stage	2nd Stage
Hemicellulosic sugars (g/l)	154.1	139.4
Xylose (g/l)	124.5	117.0
All sugars (g/l)	165.4	146.1
Formic acid (g/l)	3.01	2.93
Acetic acid (g/l)	32.4	29.7
Hydroxymethylfurfural (g/l)	nd	0.17
Furfural (g/l)	0.83	2.28
Solid		
Cellulose (%dw)	31.3	32.3
Hemicellulose (%dw)	15.8	9.12
Lignin (%dw)	48.4	53.8
Ash (%dw)	0.29	0.22

nd:

The recovery of sugars in the final liquor is 24.7% of the initial sugar content, or 43.1% of the initial xylose and only 3.7% of the initial glucose.

Concerning the recovered, unwashed solid, it can be seen that its composition has changed, its cellulose content having increased to 32.3% dw (dry weight) because very little amount of glucose in the liquor is recovered, while the content in hemicellulosic sugars decreased to 9.1% dw, approximately 20% of the initial content of the olive stone.

3.2 Fermentation of synthetic liquors

3.2.1 Fermentation of 120 g/l xylose, 10 g/l glucose synthetic liquor

As a first step in the study of microorganism performance, solutions with sugar concentrations similar to those obtained from olive stone autoclave pretreatment (120 g/l xylose, 10 g/l glucose) were prepared and fermented. No inhibitor compounds were added at this stage, while the fermentation behavior of *S. stipitis* (pH 6.5) and *E. coli* MS04 (pH 7) were compared without pH control. Under such conditions, *E. coli* MS04 was not able to convert any sugars, which is attributable to a pH decrease from 7 to 6 [17,18], while *S. stipitis* did ferment. The experiment was then repeated, but with pH control. The corresponding results are shown in Fig. 1. It can be seen that *S. stipitis* shows a shorter lag phase and fermentation started 17.5 h. By contrast, the same process by *E. coli* MS04 was three times longer. Although the highest ethanol concentration produced by *S. stipitis* (around 14 g/l) was attained after 75 h fermentation time, *E. coli* MS04 exhibited an increasing ethanol concentration, reaching 34 g/l at 120 h. The corresponding ethanol yields are 11% in the case of *S. stipitis* and 27% in the case of *E. coli* MS04. None of the microorganisms were able to convert all the sugars, in spite of allowing the process for up to 168 h. Moreover, the production of ethanol started as the uptake of glucose did. In the case of *S. stipitis*, the ethanol productivity peak was found when glucose was depleted.

3.2.2 Fermentation of 40 g/l xylose, 10 g/l glucose synthetic liquor

The described behavior could be attributed to substrate inhibition, as a relatively high sugar concentration was used. To verify this, an additional experiment with much lower sugar concentration was performed. Results are shown in Fig. 2. By comparison with Fig. 1, it can be deduced that the lag phase duration is highly dependent on the initial sugar concentration. This is more evident for *E. coli* MS04, as deduced from Fig.1a when it took more than 50 h when a high sugar concentration was used in contrast to less than 10 h in the case of lower sugar concentration (Fig.2a). Moreover, *E. coli* MS04 used up all the present sugars in about 35 h, while *S. stipitis* needed more than double the time. It is also worth noting the high ethanol yield reached by *E. coli* MS04 (50%, that is 98% of the theoretical) whereas *S. stipitis* hardly attained 30%, equivalent to only 59% of the theoretical. In this case, as it can be seen from Fig. 2d, a significant part of the substrate is converted into biomass, which is produced in a concentration up to 10 g dw/l, much higher than 1.3 g biomass/l obtained with *E. coli* MS04. Finally, Fig. 2d also shows a direct relationship between production of biomass and ethanol.

3.2.3 Effect of the initial xylose concentration on the fermentation of synthetic liquors

As a conclusion of the previous studies, it is evident that the initial concentration of xylose plays a major role on microorganism adaptation [13]. This result is in agreement with those reported with *E. coli* FBR5 [19,20], where increasing xylose concentration resulted in reduced biomass growth and decreased both xylose consumption and ethanol production. To get a deeper knowledge of this effect, a new experimental series was performed by varying the initial xylose concentration in a wider range (30, 50, 70, 110 g xylose/l, and 10 g glucose/l. *E. coli* MS04

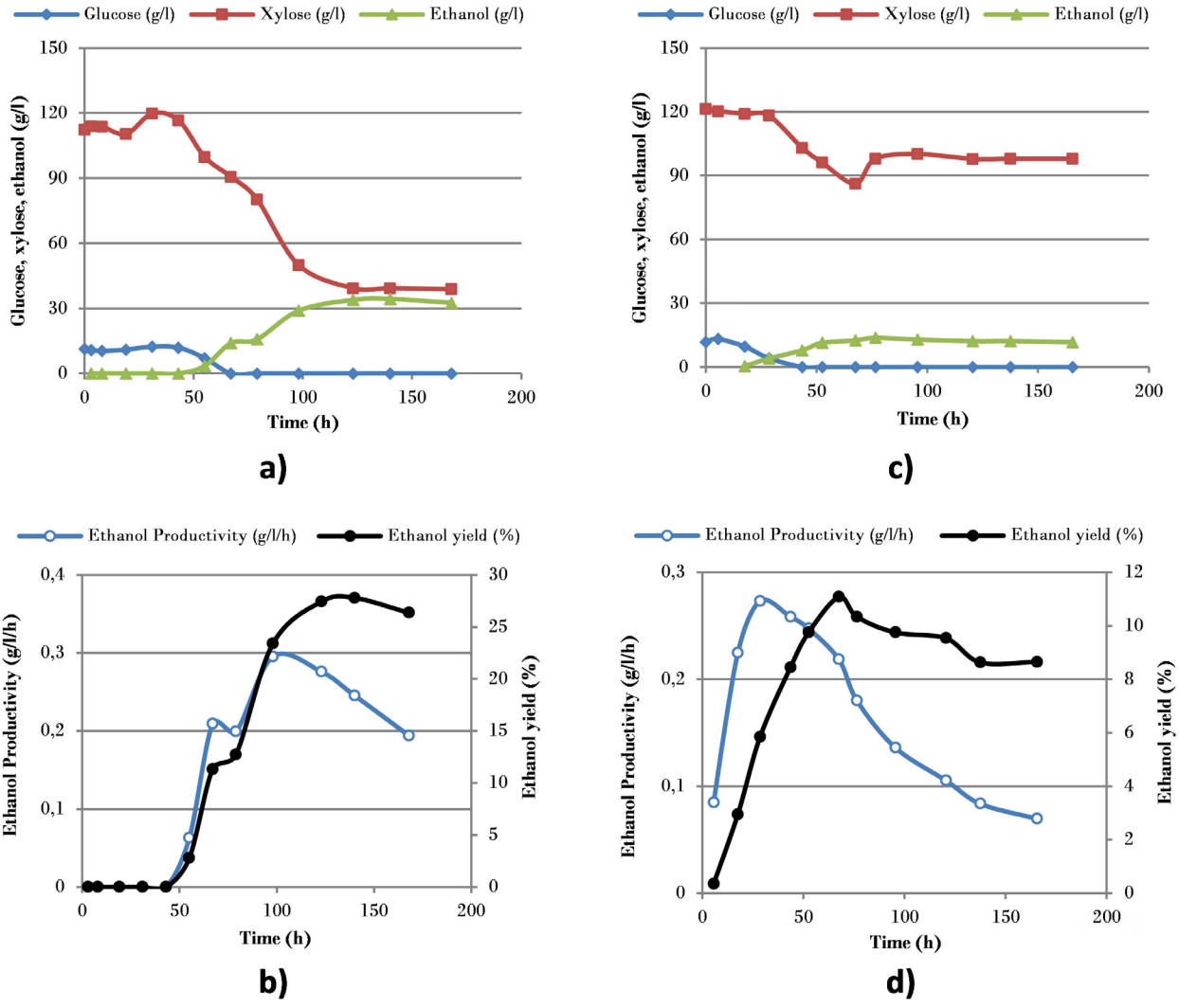


Figure 1. Time course of the fermentation of synthetic liquors at high sugar concentration, especially xylose. a),b) *E. coli* MS04 (pH 7); c),d) *S. stipitis* (pH 6.5).

was selected as the only microorganism for this study, since *S. stipitis* exhibited a comparatively lower ethanol yield as shown in the previous paragraphs.

The results from pH controlled fermentations are depicted in Fig. 3. It can be seen that increasing the initial xylose concentration produced an exponential increase in the adaptation phase.

The results show also the product inhibition effect, as in the experiments performed at the higher xylose concentration, the ethanol concentration reached up to 35 g/l while there was still xylose left in the medium (glucose was completely exhausted in all cases). Product inhibition has also been previously reported. For example, *E. coli* LY01 showed inhibition above 45 g ethanol/l, even if this strain was able to produce up to 60 g ethanol/l from 140 g xylose/l, being the highest value reached so far by *E. coli* [9]

The ethanol yield decreased linearly as long as the xylose concentration increased for initial concentrations greater than 50 g xylose/l. For initial xylose concentrations below 35 g/l, the ethanol yield is close to 0.5 g ethanol/g total sugar, which is equivalent to 99% of the theoretical value. Regarding the ethanol volumetric productivity, it can be seen that the highest values are obtained at the lowest assayed xylose concentrations. Finally, the final biomass concentrations at the end of the experiments are approximately the same, 1.5 g dw biomass/l, regardless the initial xylose concentration.

3.2.4 Fermentation of synthetic liquors including inhibitors

The next step in our study was to assess the fermentation in the presence of inhibitors. Two experiments were performed using in the first case 5 g glucose/l and 63 g xylose/l, which is approximately half the concentrations of these sugars obtained after autoclave treatment, and in the second case a higher glucose concentration (10 g glucose/l and 50 g xylose/l). Both experiments were repeated with the addition of acetic acid, formic acid and furfural at half the concentration detected in the autoclave liquors, e.g., 17, 1.6 and 0.8 g/l, respectively. The main fermentation results obtained using *E. coli* MS04 in pH controlled cultures (pH 7) are shown in Fig.4. Big differences can be seen between both types of experiments –with and without inhibitors. Concerning ethanol production, this is much lower in the experiments with added inhibitors, reaching only 20 g/l after 144 h fermentation time and an almost constant 0.2 g ethanol/l/h volumetric productivity, Fig. 4c. By contrast, the experiments without added inhibitors produced up to 30 g ethanol/l in the third of the time. The ethanol yield was also decreased by 30% because the xylose was not completely consumed after 144 h. In accordance with the above mentioned results, the biomass production was also much higher in the case of no inhibitor added fermentations (>1.5 g/l versus <1 g/l). The microorganism behavior as a function of the inhibitors is independent of the sugar concentration. Furfural was totally consumed (0.8 g/l in 18 h) and then the production of ethanol took place, which agrees with reports by Nieves et al. [21]. Concerning the other inhibitors, a small amount of formic acid is also consumed, while acetic acid did not change throughout the 144 h of fermentation experiment (see Fig. 5).

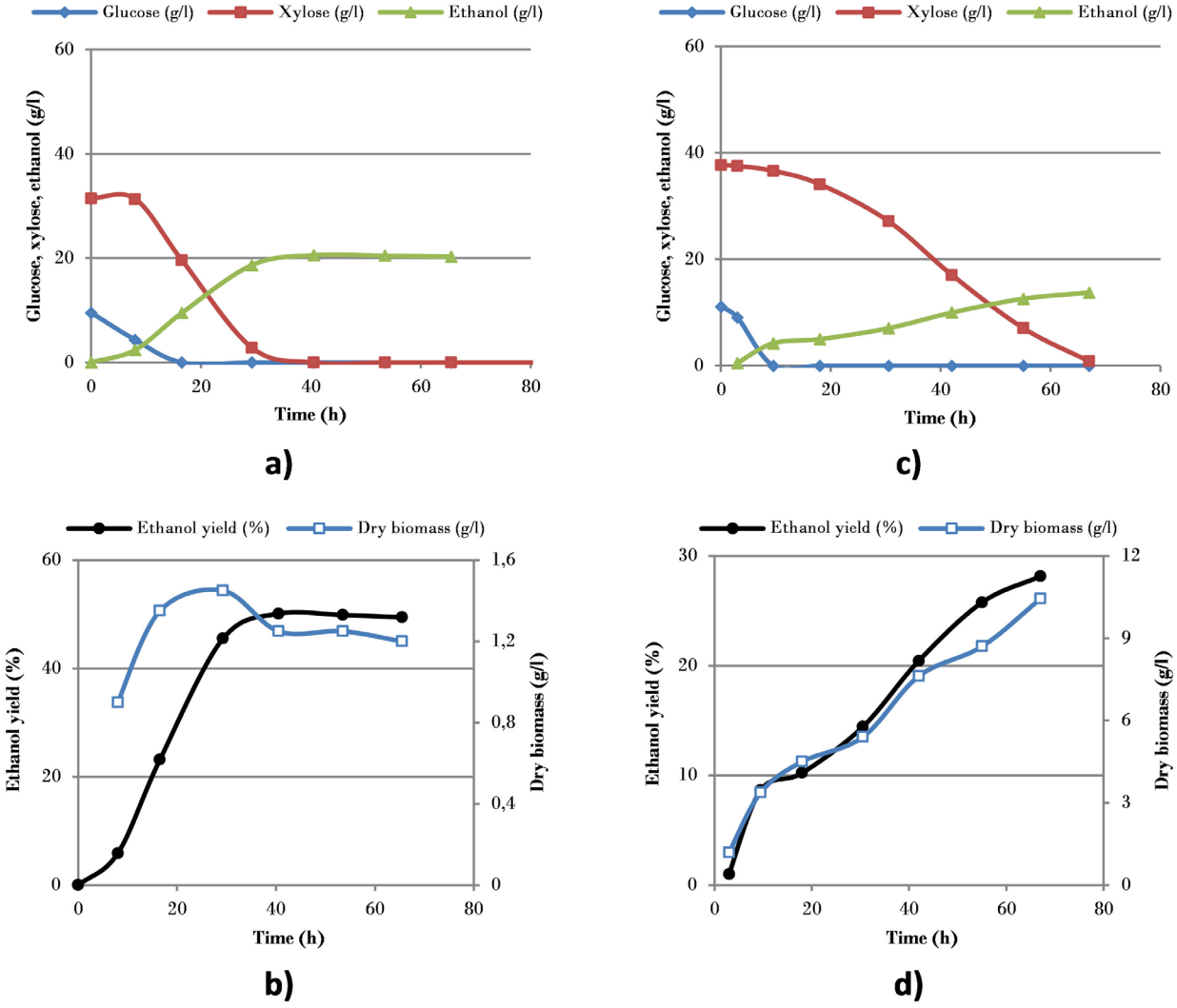


Figure 2. Time course of the main fermentation parameters for a low sugar content (40 g xylose/l, 10 g glucose/l) synthetic liquor. Microorganisms *E. coli* MS04 (a,b) (pH 7), *S. stipitis* (c,d) (pH 6.5).

3.2.5 Influence of pH on the fermentation of synthetic liquors

As a conclusion of the previous sections, it was clear that *E. coli* MS04 was a better option than *S. stipitis* for producing ethanol from synthetic liquors similar to those obtained from autoclave pretreated olive stones. As a next step, the comparison between *E. coli* MS04 and *E. coli* MM160, another xylose fermenting, well-documented microorganism was investigated. Firstly, the influence of pH in the fermentation behavior of synthetic liquors without inhibitors was assessed to determine whether 6.5 or 7 was the optimal pH for *E. coli* MM160 when used on olive stone liquors, based on literature reports [10,18]. Available reports state [22] that the optimal pH range for ethanol production by ethanologenic *E. coli* on either glucose or xylose was 6.0-6.5, while the highest value of the maximum specific growth rate (μ_m) was obtained at pH 7.0.

The initial concentration of sugars was established in 48 g xylose/l and 10 g glucose/l. The fermentation results are shown in Fig. 6. It can be seen that the ethanol productivity was higher for *E. coli* MS04, reaching the maximum ethanol concentration of 29 g/l in some 30 h while *E. coli* MM160 required 50 h regardless of the initial pH. The ethanol yield was near 50% in the three experiments, and sugars were completely consumed. Regarding biomass production, little differences were observed between both microorganisms, Fig. 6d.

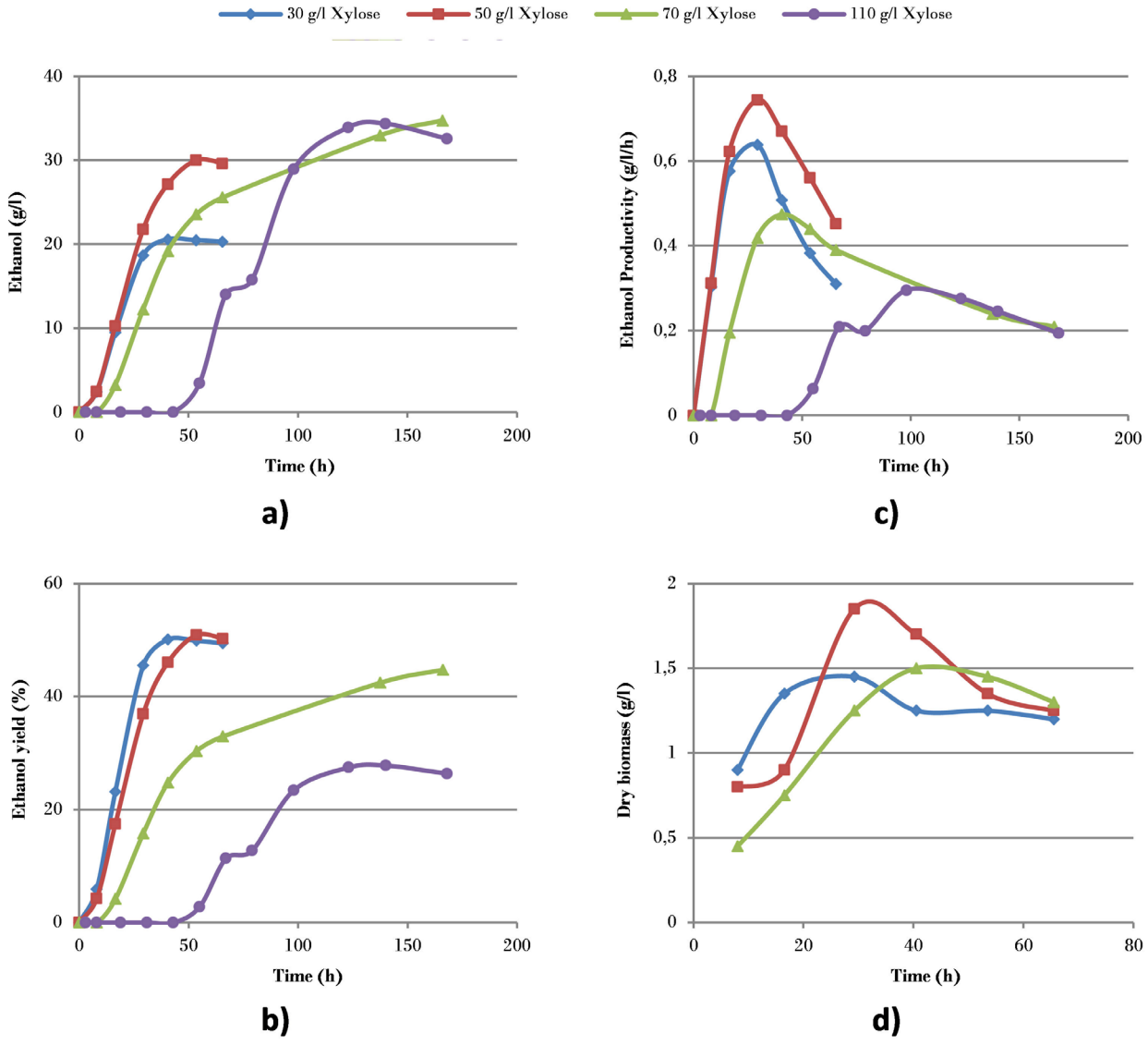


Figure 3. Time course of the main fermentation parameters obtained from controlled pH (pH 7) cultures of *E. coli* MS04 at different initial xylose concentrations (glucose concentration in all cases was 10 g/l. Experiments were performed in triplicate and standard errors were <5%.

3.2.6 Comparison of fermentation by *E. coli* strains on synthetic liquor with added inhibitors

As a final step of the study on synthetic liquors, both the *E. coli* strains were assayed using a liquor containing 58 g xylose/l, 6 g glucose/l, 17.5 g acetic acid/l, 1.8 g formic acid/l and 0.5 g furfural/l, which are approximately half the concentration found in the autoclave liquid fractions. In addition, *E. coli* MM160 was tested at both 6.5 and 7 initial pH. Results are shown in Fig. 7. Basically, *E. coli* MM160 presented higher volumetric ethanol productivity at pH 7, reaching 30 g ethanol/l after 100 h fermentation time, while it required more than 150 h at pH 6.5. Concerning *E. coli* MS04, the highest ethanol production was attained after 125 h. The ethanol yield was greater than 46% in all cases, with up to 50% in the case of *E. coli* MS04. Regarding biomass production, the highest value was about 1 g/l for *E. coli* MM160 at pH 7. It is also worth noting a direct relationship between the consumption of furfural and the initial pH. In experiments at pH 7, furfural was completely depleted in 10 h, while it took more than 25 h in the experiment performed at 6.5 initial pH (Fig. 7e).

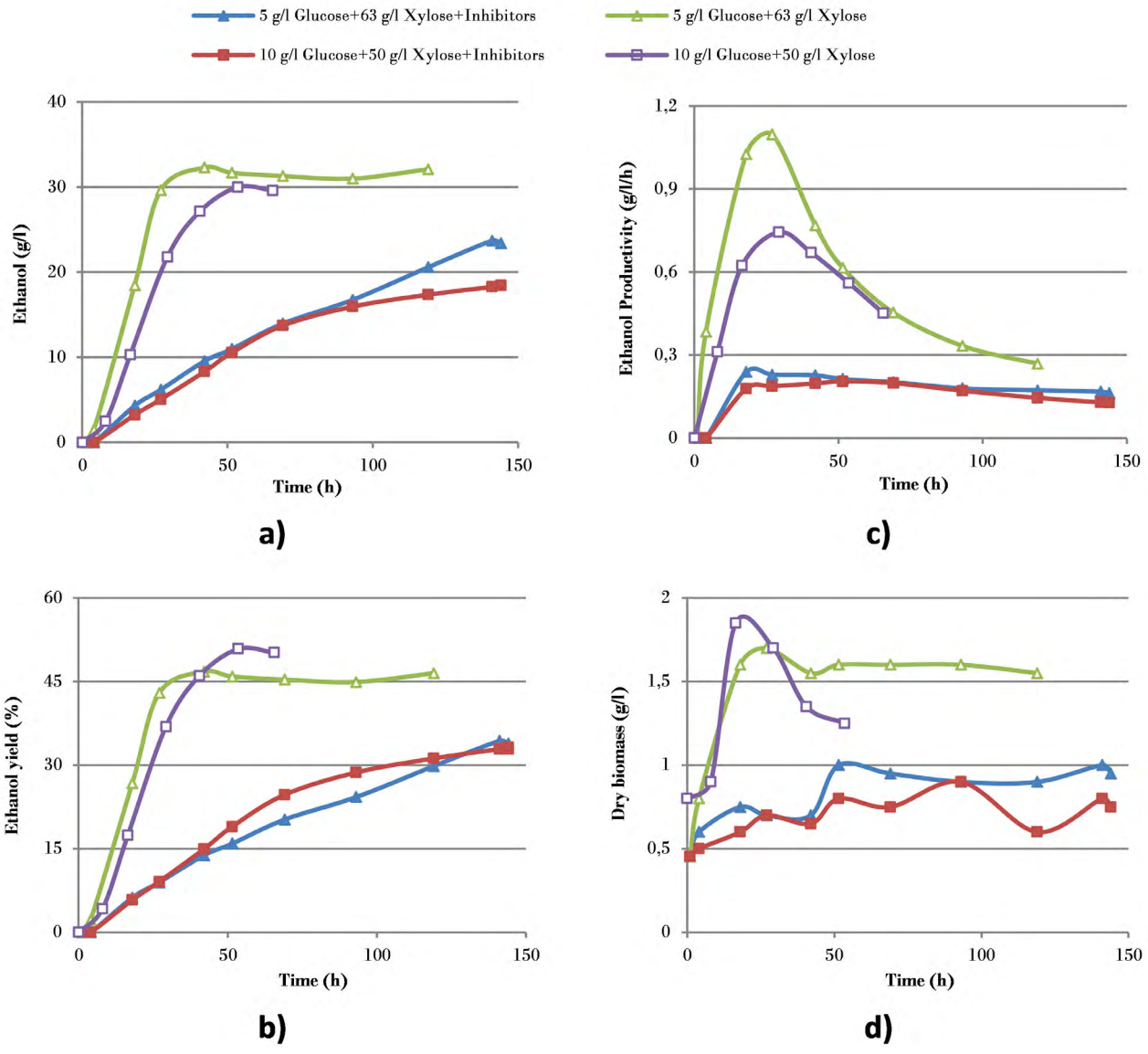


Figure 4. Time course of the main fermentation parameters obtained from controlled pH cultures (pH 7) of *E. coli* MS04 at different initial sugar concentrations with (bold symbols) and without (hollow symbols) added inhibitors (17 g acetic acid/l, 1.6 g formic acid/l and 0.8 g furfural/l). Experiments were performed in triplicate and standard errors were <5%.

3.3 Fermentation of olive stone real liquors

Considering the results attained so far the study of the fermentation on the real liquor obtained by autoclave treatment of olive stones was performed by *E. coli* MM160 (pH 7) because of its higher ethanol productivity and ethanol yield. Considering that sugar concentrations greater than 120 g/l would result in a slow fermentation process, liquors issued from autoclave were diluted by 50%. Both untreated and detoxified liquors were assayed. Table 2 shows the composition of synthetic, and untreated and detoxified real liquors. As can be seen they are quite similar, except for furfural composition, which is reduced by two thirds when the liquor is detoxified.

The main fermentation results are shown in Fig. 8 where great differences among synthetic, untreated real and detoxified real liquors are evidenced. *E. coli* MM160

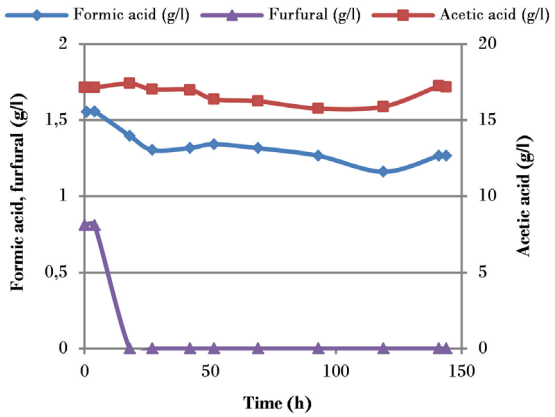


Figure 5. Time evolution of the concentration of added inhibitors obtained from controlled pH cultures of *E. coli* MS04 (glucose concentration 5 g/l; xylose concentration, 63 g/l). Experiments were performed in triplicate and standard errors were <5%.

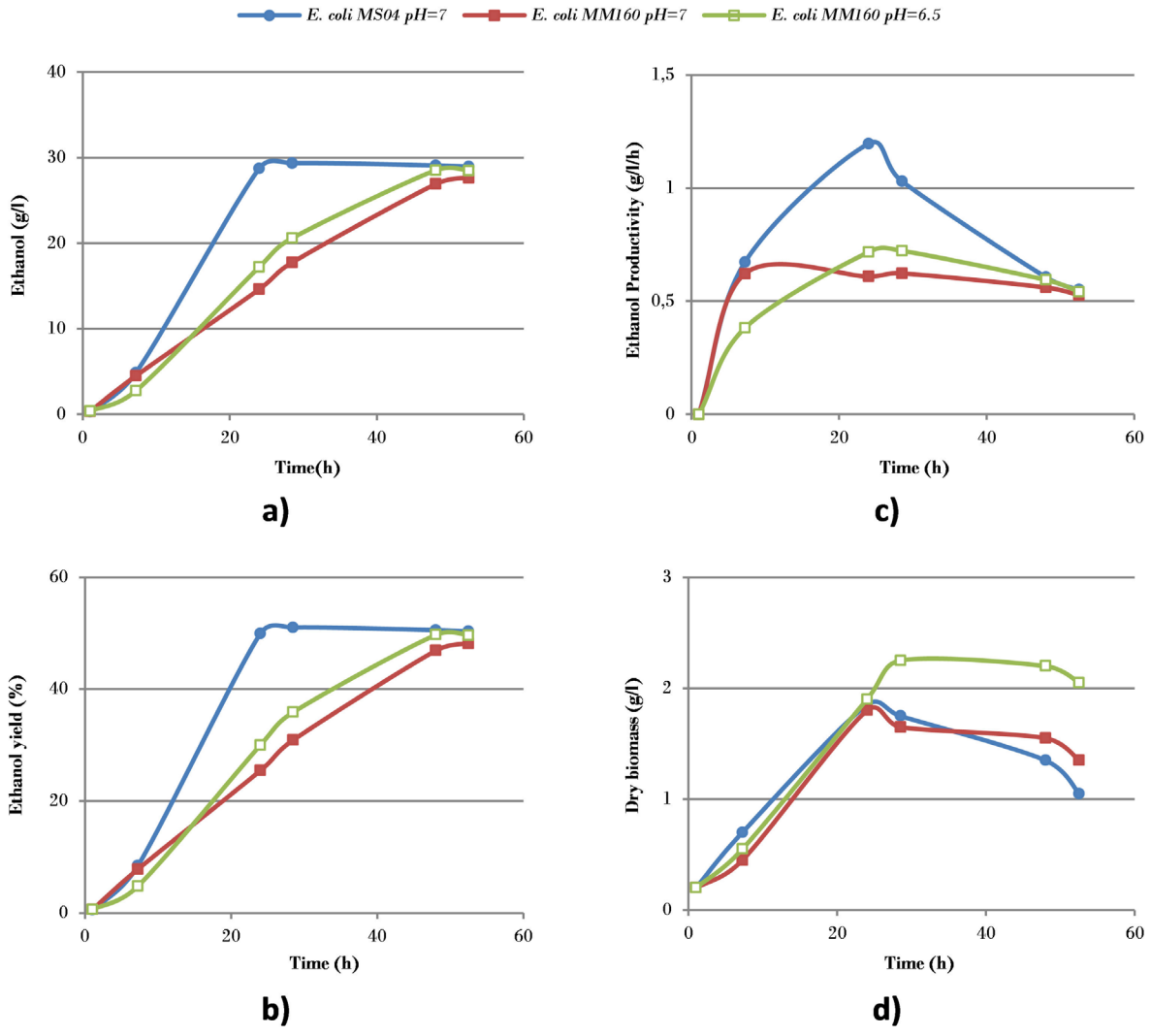


Figure 6. Time course of the main fermentation parameters obtained from controlled pH cultures of *E. coli* MS04 and *E. coli* MM160 at different initial pH and 48 g xylose/l and 10 g glucose/l. Experiments were performed in triplicate and standard errors were <5%.

Table 2. Comparison of composition of liquors used for fermentation

	Real liquor	Detoxified real liquor	Synthetic liquor
Total sugars	74.5	76.9	64.8
Glucose (g/l)	3.8	4.8	5.9
Xylose (g/l)	62.3	63.2	58.5
Acetic acid (g/l)	18.2	20.2	17.5
Formic acid (g/l)	0.8	1.1	1.8
Furfural (g/l)	0.6	0.2	0.5

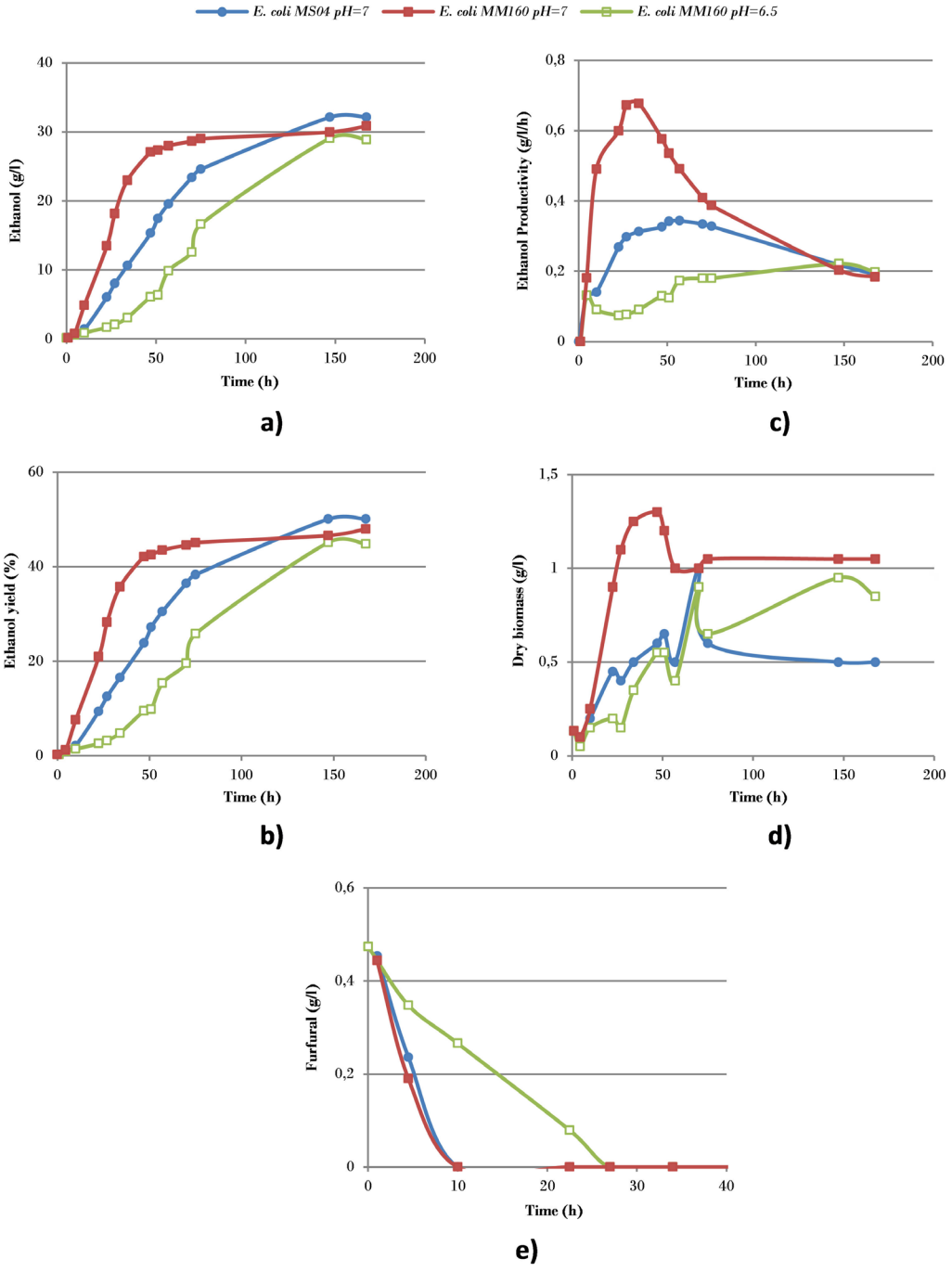


Figure 7. Comparison of fermentation experiments of controlled pH cultures with *E. coli* MS04 and *E. coli* MM160 using synthetic liquors composed of 58 g xylose/l, 6 g glucose/l, 17.5 g acetic acid/l, 1.8 g formic acid/l and 0.5 g furfural/l. Experiments were performed in triplicate and standard errors were <5%.

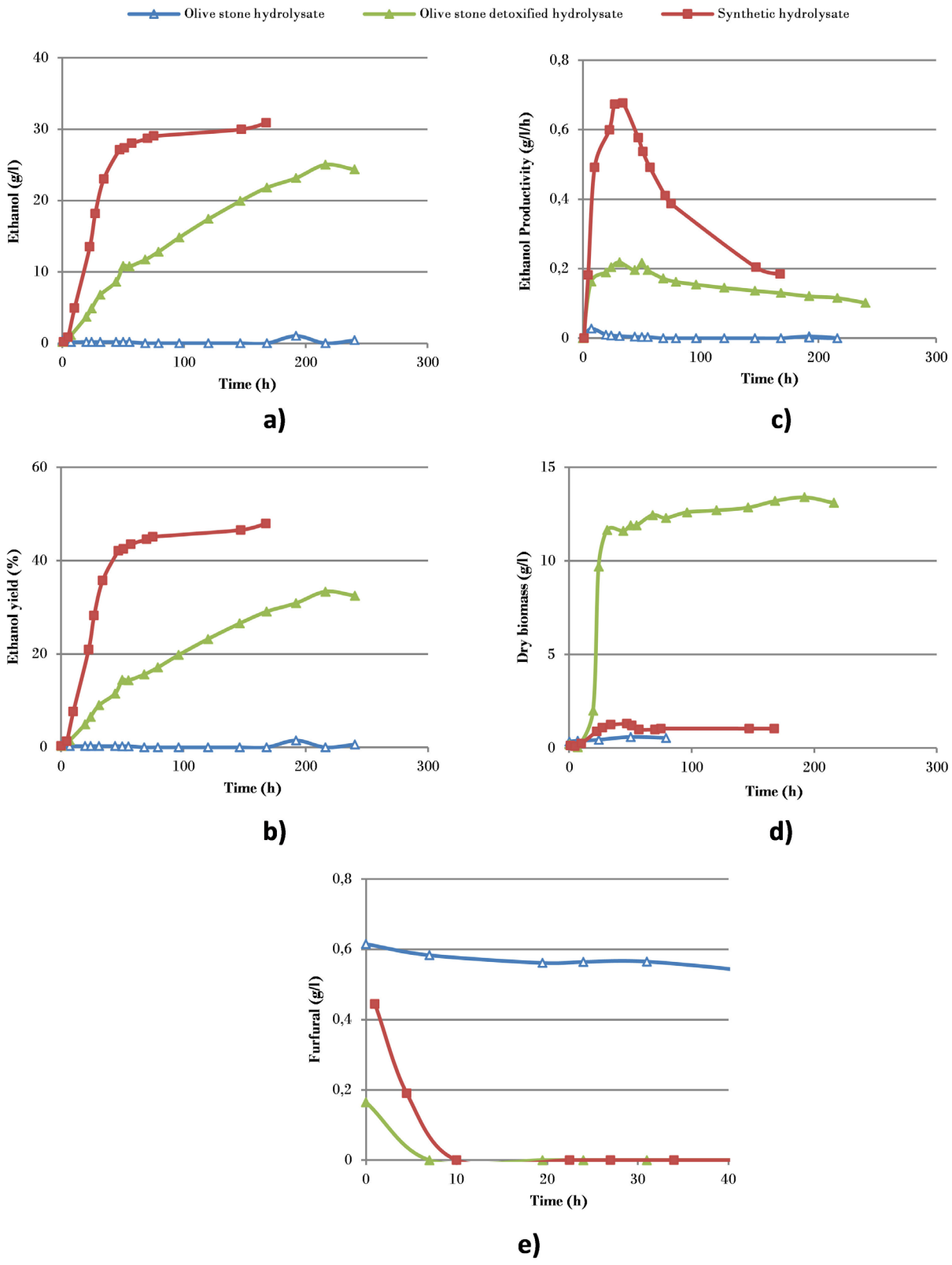


Figure 8. Comparison of fermentation parameters evolution for controlled pH cultures (pH 7) with *E. coli* MM160 using synthetic, real untreated and real detoxified olive stone liquors. Real liquors were dilute by 50%. Experiments were performed in triplicate and standard errors were <5%.

was not able to ferment the untreated real liquor. It was able to ferment the detoxified real liquor, but much more slowly than the synthetic liquor. This seems to point out that the considered inhibitor compounds (acetic acid, formic acid and furfural) are not the most influential compounds, but that there must be other compounds such as polyphenols that are partially eliminated during the detoxification and that also have inhibitory effect [23,24]. The highest ethanol concentration was 25 g/l, attained at 216 h, while the volumetric productivity was under 0.2 g ethanol/l/h. The highest ethanol yield was 35%, equivalent to 69% of the theoretical. In relative terms, this is a high value if it is considered that the starting point is a sugar concentrated liquor, more than 75 g total sugars/l, and with the presence of inhibitory compounds. It is also worth noting that on the one side the microorganism was not able of consuming all the sugars. And on the other side a biomass concentration as high as 13 g dw/l was produced, Fig. 8d; by contrast, none of the previous experiments with synthetic liquors resulted in a final biomass concentration greater than 1.5 g dw/l.

E. coli MS04 was also assayed for fermentation of real liquors. As in the case of *E. coli* MM160, a high biomass concentration greater than 12.5 g dw/l was obtained, confirming that detoxification enhanced biomass production, Fig. 9. Concerning furfural in the real detoxified liquor, once this compound was consumed the production of ethanol took place, while no furfural consumption was detected when using the undetoxified real liquor, Fig. 8e.

The ethanol concentration obtained from the real hydrolysates of olive stones (25 g ethanol/l) compares favourably with other results on the same or similar feedstocks. For example, using *Pachysolen tannophilus*, another xylose-fermenting yeast, only 2 g ethanol/l were achieved after 150 h fermentation time [25] and 3.5 g ethanol/l after 300 h when olive tree pruning biomass was used as a substrate [26]. The best result on olive tree biomass was found when *S. stipitis* was

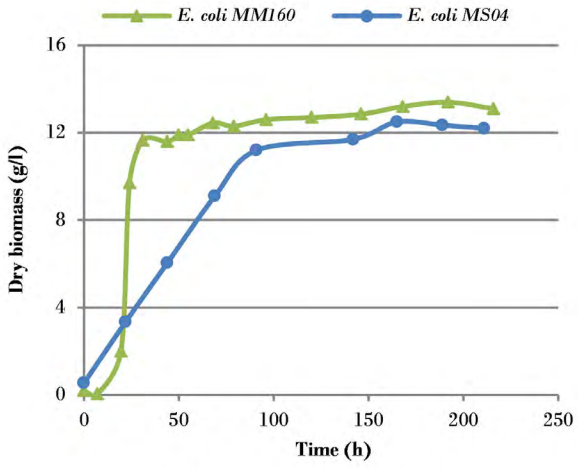


Figure 9. Biomass production during the fermentation of diluted real detoxified liquor by *E. coli* MM160 and *E. coli* MS04 (both pH 7).

used for hemicellulosic hydrolysates fermentation, and 9 g ethanol/l concentration was attained [6]. Concernig *E. coli* results, olive pomace hydrolysate fermentation resulted in 8.1 g ethanol/l [13] which is still a low concentration for distillation purposes.

Comparing with other raw materials, *E. coli* KO11 produced up to 30 g ethanol/l from sugarcane bagasse hydrolysates after 125 h fermentation time [27], although acetic acid concentrations were lower than 7 g/l instead of 20 g/l found in our hydrolysates. Also Saha et al. reported 21 g ethanol/l from the fermentation during 120 h of wheat straw hydrolysates using *E. coli* FBR5, and 2.5 g acetic acid/l in the medium [17]. *E. coli* KO11 also produced 19.4 g ethanol/l at 144 h fermentation of corn stover hydrolysates [28].

Considering the above mentioned results from different raw materials, the ethanol concentration reported in this article from olive stone hydrolysates is among the higher ones, although requiring longer fermentation time due to higher concentrations of acetic acid. Table 3 shows a comparison of ethanol yields produced with different strains of ethanologenic *E. coli*.

Table 3. Results of ethanol production from fermentation of different hydrolysates using ethanologenic *E. coli*.

Raw material	Microorganism	Ethanol yield (g/g raw material)	Reference
Sugarcane bagasse	*E. coli* MM160	0.21	[10]
Sugarcane bagasse	*E. coli* MM170	0.27	[29]
Wheat straw	*E. coli* FBR5	0.31	[17]
Corn stover	*E. coli* FBR5	0.31	[30]
Eucalyptus benthamii	*E. coli* SL100	0.24	[11]
Olive stones	*E. coli* MM160	0.26*	This work

*0.054 g ethanol/g olive stone (this work) + 0.206 g/g (ethanol potentially obtainable if all remaining sugars were converted).

3.4 Production of bioethanol from highly concentrated hemicellulosic liquors of olive stones. Mass balance of the proposed scheme.

Considering all the findings derived from the present investigation, Fig. 10 summarizes a possible process for the production of bioethanol from olive stones. The corresponding mass balance at each process step is also shown. The first step of the process is a low-energy demanding (130°C) dilute acid autoclave pretreatment which results in a highly sugar-concentrated liquor, which is separated from a cellulose-enriched solid. Based on 100 g dry matter olive stones, the hydrolysate accounts for 20 g dm being constituted by a mix of sugars, acetic acid and other minor components. Xylose accounts for up to 80% of the total sugars in the liquor, with a total concentration of more than 140 g hemicellulosic sugars/l. The fermentation of this liquor requires firstly dilution by 50% and a detoxification step by overliming. Out of three xylose fermenting microorganisms assayed, *E. coli* MM160 has proved to be best one according to the liquor features. After 216 h fermentation time an ethanol solution of 25 g/l is obtained, while some unfermented sugars remain in the broth. Referred to the initial feedstock, 5.4 g ethanol can be produced from 100 g olive stones, which can be an important help for the economical efficiency of the global conversion under a biorefinery scheme.

Preliminary tecno-economic studies are on the way using the commercial software Aspen Plus® v8.0. These studies are considering on the one side the ethanol production from the hemicellulosic hydrolysate according to the description in Fig. 10. On the other side, the ongoing studies take into account also the ethanol that can be produced from the remaining sugars in the olives tones, once the hemicellulosic sugars have been considered. In this case, the kinetics models proposed by Girisuta et al. [31] for the cellulose hydrolysis were adopted for the dilute acid pretreatment of olive stone; then SSCF (Simultaneous Saccharification and CoFermentation) was considered through the kinetic models proposed by Morales-Rodríguez [32]. MATLAB was used for the resolution of kinetic equations. Globally, the simulation with Aspen Plus® revealed that 1018 kg ethanol/h can be produced from 10,000 kg olive stones/h, with 99.8% purity after distillation and molecular sieves treatment. As far as the preliminary economic evaluation is concerned, the cost of

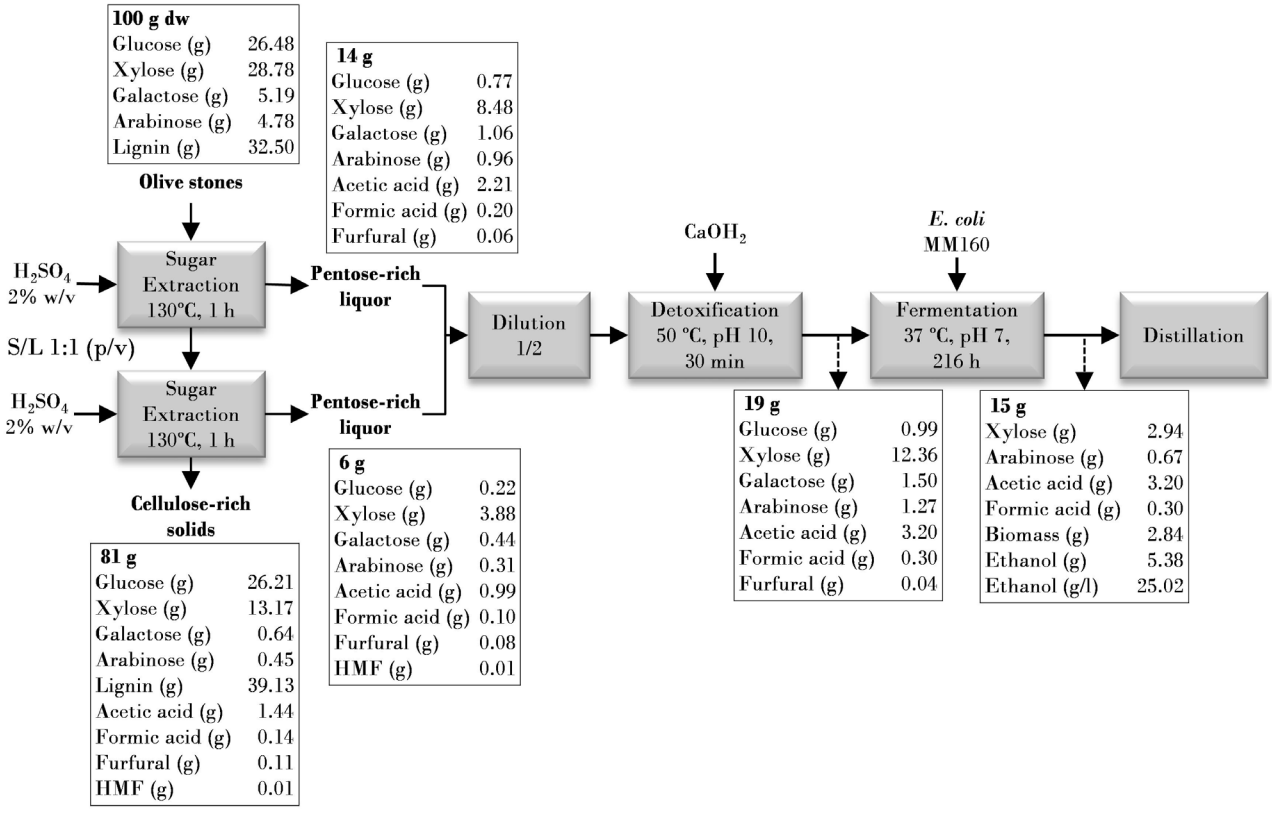


Figure 10. Mass balance for the production of ethanol from hemicellulosic liquors obtained by dilute acid autoclave pretreatment of olive stones and fermentation by *E. coli* MM160.

producing one liter of bioethanol from olive stones in the above described conditions is 1.67 US$.

4 Conclusions

Hemicellulosic sugars can be separated from the lignocellulosic matrix of olive stones by a procedure consisting in double dilute acid pretreatment at mild temperature. The use of a liquid to solid ratio of 2 is enough to ensure that the liquors obtained form a concentrated sugar solution containing 40% of the initial hemicellulosic sugars present in the raw material, with xylose being the major sugar present. *E. coli* MM160 is the best performing microorganism of the three ones assayed. The hydrolysate could not be directly fermented but it required a detoxification step by overliming. The resulting liquor, containing 80 g/l of sugars (80% xylose), 20 g/l acetic acid, 1.2 g/l formic acid and 0.2 g/l furfural was converted into 25 g/l ethanol with 35% ethanol yield or 70% theoretical ethanol yield. The production of biomass in the fermentation of real hydrolysate is much greater than that obtained in synthetic media, which can be attributed to the presence of other non-determined compounds in the hydrolysates.

The remaining solids, consisting basically in cellulose and lignin, can be submitted to another pretreatment step to make them accessible to enzymatic hydrolysis for glucose release. Further research is needed to establish appropriate conditions for this second step.

As a general conclusion, this work demonstrated that olive stones can be used as feedstock for ethanol production under a biorefinery scheme, and *E. coli* MM160 is a useful microorganism for taking full advantage of the hemicellulosic sugar content of the olive stones.

Olive stones constitute a raw material with a potential for producing bioetanol similar to others like sugarcane bagasse, wheat straw or corn stover. The preliminary production cost obtained by techno-economic simulation may indicate that this material deserves further studies to improve yields and productivity and lower prices.

Further research work will be focused on the pretreated solids conversion by means of enzymatic hydrolysis and fermentation, and process integration to develop an efficient biorefinery based on olive stones.

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